01

QUALITIES OF A COLOUR TELEVISION/ VIDEO CAMERA TECHNICIAN

1. A good CTV/video Camera technician must be fully acquated with the theory of colour metric, the thri-colour Kinescope and accessory and the assortment of colour circuit.

2. He Should note that diagnosing CTV trouble is an excercise in training, logic, patience, and the use of the senses, books magazine articles. Manufacturer Service notes are available as a Source of Knowledge.

3. Self displine is needed to take advantage of the information on the regular basis.

4. The experienced technician should learn often the headway that he must look, listen, smell and feel before he leaps to conclutions.

5. He should note that CTV is more complex to service than its black & white counterpart.

6. The experienced Service man have to check overall receiver performance first on the basis of the receiving as a black and white set, only when the technicain is satisfied that the b/w TV performance is satisfactory, will he view a colour picture.

7. He will then note whether colour are true and remain locked in Sync.

8. The failure in Colour are carefully analysed using all senses, burning odor indicating an overheated resistor, a tiny visible crack in the [illegible]

VERTICAL SYNC PROBLEMS.

The lack of vertical sync or a very critical vertical sync can be caused by improper adjustment of controls, as well as by defective circuits.

1. The first step in troubleshooting this symptom is to adjust the vertical hold (sometimes called vertical lock or vertical sync) control.

There is no problem if vertical sync can be obtained

2 with the vertical hold control at mid range.

3 If the vertical hold control must be full or (near full) or it is necessary to readjust the control frequently, there is a problem in the vertical sync circuits.

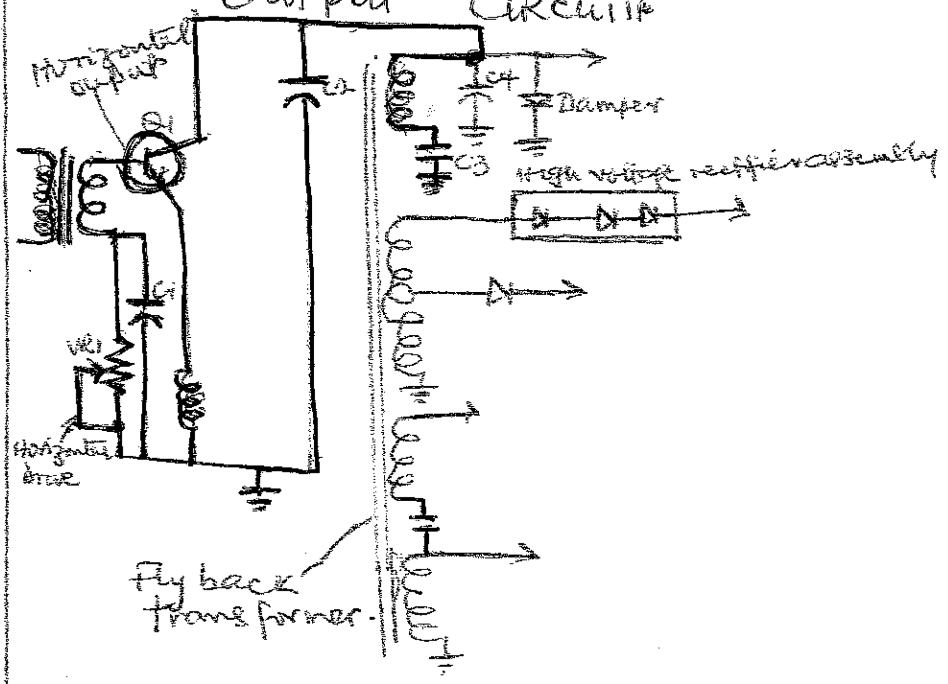
4 If the problem cannot be cured by adjustment, check for the presence of proper sync pulses at the input to the vertical sweep circuit at the primary of the transformer.

5 Then check the sawtooth sweep portion of the vertical oscillator at the base of a first transistor. The vertical pulse will appear to ride on the sawtooth.

6. If the sync pulses are normal and it is impossible to get proper vertical sync, the most likely causes are:

Leaking by pass capacitors C5, C6 in the vertical sweep circuit.
Worn vertical hold control.
Leaking or defective vertical oscillator transistor.
Defective vertical transformer (leakage winding or some similar imaginal defect).

HIGH VOLTAGE AND HORIZONTAL OUTPUT CIRCUIT ~~From~~ 65—85 TEST



HIGH VOLTAGE AND HORIZONTAL OUTPUT CIRCUITS (ALL SOLID STATE TYPE).

The function of the above circuit is to....

1. Provide a high voltage for the picture-tube second anode
2. Produce a horizontal deflection voltage (horizontal sweep) to the picture-tube deflection yoke.
3. Practically, the circuit also supply a boost voltage for the picture tube focus and accelerating grids.
4. Supply a voltage source for the video output transistor.
5. In some cases, the horizontal output circuit also supply an AFC signal to control the frequency of the horizontal oscillator and an AGC signal to provide keyed AGC.

RECOMMENDED TROUBLE SHOOTING APPROACH.

Many trouble symptoms caused by horizontal output circuits can also be caused by defects in other cts. ~~Example~~: Dark screen ~~and~~ (no raster) or insufficient width are two good examples.

If the low-voltage Power Supply is completely inoperate (or almost zero) the screen can be dark. If the low-voltage supply is low, the picture width can be decrease. Of course in either case, the sound is absent or abnormal. On the other hand, if the horizontal oscillati and drive ccts are defective, there will be no high voltage or horizontal Sweep.

If either high voltage or the boost voltage is absent or abnormal, this will isolate the trouble ~~to~~ the corresponding cct. If both voltages are absent or absent (with the drive signal good) the problem is in horizontal output transistor or related cct part like capacitor, flyback transformer etc.

If the high voltage is absent, check for ac at the anode side of the high voltage rectifier. Observe all the usual precautions when measuring high voltage. In addition do not make arc test of the high voltage with solid circuits. The transient voltages can damage the transistors.

DARK SCREEN BUT SOUND NORMAL.

It is likely that the following(s) is/are responsible for the above fault.

1. horizontal output
2. High voltage circuit
3. Picture tube can also be defective.
4. Lack of filament voltage
5. Open, shorted or leaking capacitors
6. Leaking horizontal output transistor
7. Open or shorted diodes
8. Defective flyback transformer.

~~PICTURE~~ OVERSCAN.

The picture becomes dim even with the brightness fully on, and there is abnormal enlargement of the picture or raster.

(1) In some circuits, the brightness control operates in reverse, rotating the control for an increase in brightness produces a decrease.

(2) If the picture overscan is not accompanied by sufficient width (a symptom of horizontal output failure) the cause is a failure in the high voltage circuit only.

(3) Any circuit problem that reduces (but does not completely eliminates) the high voltage output to the picture tube second anode can cause overscan.

The general likely defects are ~~leaking~~

1. Leaking high voltage capacitor.
2. Defective high voltage tube and rectifiers
3. Leaking high voltage leads
4. Defect in high voltage circuit protective resistance (if any)
5. Shorted turns in high voltage winding of the flyback transformer.
6. If the high voltage is low, check any high voltage filter capacitors for leakage.
7. Many colour television ~~produces~~ pictures and raster bloom when both brightness and contrasts controls are turned fully up. This is an inherent design problem (lack of high voltage regulation).

COLOUR CONVERGENCE

Poor Purity, improper convergence adjustments, a shift in the ~~grey~~ gray scale tracking and other colour section defects permits colour ~~to~~ appear on the receiver screen during monochrome reception.

To clear this critical adjustment, the technician must examine the entire screen to observe the effect of each control as the changes the setting.

The difficulty and inconvinence of the task if station programme material is used for observation, can be easily imagined, why the technician is attempting to concentrate on specific areas the picture is changing in contact, brightness and contrast.

Pattern generator with stable fixed pattern in black and white is best to be used for convergence because colour fridging is more noticable on black and white programme.

Finally colour fridging is most apparent when mostly of the screen is dark and only a small percentage is illuminated with dots or bars.